

Quantum Teleportation: Protocol, Implementation, and Applications

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Abstract

This report provides a comprehensive study of quantum teleportation, covering its protocol, implementation, and applications. We begin with an introduction to the fundamental concepts and historical development of quantum teleportation. We then present the theoretical framework and the step-by-step protocol for achieving quantum teleportation. Following this, we discuss various implementation strategies and their practical challenges. Finally, we explore the current and potential applications of quantum teleportation in quantum information processing and communication.

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I. INTRODUCTION

II. MATHEMATICAL BACKGROUND

A. Qubits and the Hilbert Space Formalism

The fundamental unit of quantum information is the "qubit". Unlike classical bit, which takes the value of either 0 or 1 at one of the time, a qubit exists in a two-dimensional Hilbert space $\mathcal{H} = \text{span}\{|0\rangle, |1\rangle\} \cong \mathbb{C}^2$, and its state can be described by a unit vector with an orthonormal computational basis $\{|0\rangle, |1\rangle\}$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1 \quad (1)$$

where α and β are the probability amplitudes: measurement yielding $|0\rangle$ or $|1\rangle$ has a probability $|\alpha|^2$ and $|\beta|^2$, respectively [1]. Unlike classical bit, a qubit can exist in a superposition of both states, which has no classical analogue. Physical realization of qubits include the two polarization of photon and two energy levels of an electron in an atom.

Bloch Sphere As an phase factor $e^{i\gamma}$ does not have observable effect, Eq. 1 can be written in a more general expression

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle, \quad \theta \in [0, \pi], \quad \varphi \in [0, 2\pi) \quad (2)$$

The two parameters θ and φ define a unique point on the unit three-dimensional sphere, known as the Bloch sphere as shown in Figure 1. The north pole ($\theta = 0$) corresponds to $|0\rangle$, while the south pole corresponds to $|1\rangle$, and every point on the equator represents an equal superposition such as $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$.

Multi-qubit systems. For a system of two qubits, we use the tensor product of the individual Hilbert space $\mathcal{H}_{12} = \mathcal{H}_1 \otimes \mathcal{H}_2 \cong \mathbb{C}^4$ to represent the state space, with computational basis kets $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$. Here we use the ordering convention. That is, $|01\rangle = |0\rangle_1 \otimes |1\rangle_1$. Similarly, if it is a system of three qubits, then $|010\rangle = |0\rangle_1 \otimes |1\rangle_2 \otimes |0\rangle_3$. A general two-qubit state takes the form

$$|\psi\rangle = \alpha_{00} |00\rangle + \alpha_{01} |01\rangle + \alpha_{10} |10\rangle + \alpha_{11} |11\rangle, \quad \sum_x |\alpha_x|^2 = 1 \quad (3)$$

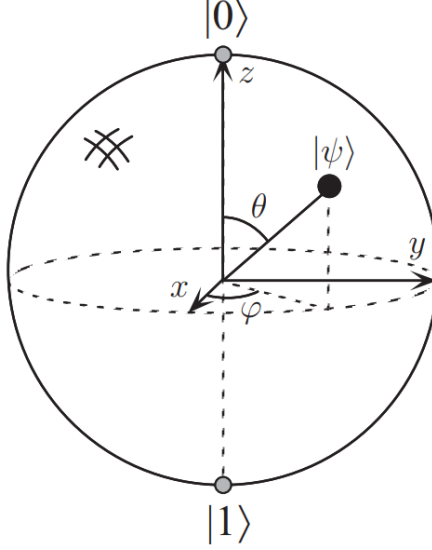


FIG. 1: Bloch sphere schematic. An arbitrary qubit can be represented by one point on the sphere [1].

B. Entanglement and the Bell States

A two-qubit state is called separable if it can be factored as $|\psi\phi\rangle = |\psi\rangle_A \otimes |\phi\rangle_B$. This means that qubit A is in state $|\psi\rangle$, while qubit B is in state $|\phi\rangle$. On the other hand, an entangled state can not be factorized into the product of two single-qubit states. In this case, the quantum state can only be understood as a property of the whole system. The most important entangled states for quantum teleportation are the Bell states [1, 2]

$$\begin{aligned} |\Phi^+\rangle &= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), & |\Phi^-\rangle &= \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \\ |\Psi^+\rangle &= \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle), & |\Psi^-\rangle &= \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle). \end{aligned} \quad (4)$$

These four states form an orthonormal basis for the two-qubit Hilbert space. Besides, they are maximally entangled: tracing out either qubit leaves the reduced density matrix $\rho = I/2$, where I is the identity matrix. For example, consider the Bell state $|\Phi^+\rangle_{AB}$, then the corresponding density matrix is $\rho_{AB} = |\Phi^+\rangle\langle\Phi^+|$. Now, if we only look at the qubit A subsystem, we trace out qubit B to get the reduced density matrix as

$$\rho_A = \text{tr}_B(\rho_{AB}) = \frac{I_A}{2}. \quad (5)$$

Similarly, we have $\rho_B = \frac{I_B}{2}$. This implies that when measuring either qubit, we get equal probability of getting $|0\rangle$ or $|1\rangle$. However, the measurement results of system A and system B are perfectly correlated. For example, if system A measure the entangled state and gets $|0\rangle$, then B gets $|0\rangle$. Similarly, if A obtains $|1\rangle$, then B gets $|1\rangle$. That is, each local measurement result is random, but it becomes highly correlated once we look at the system as a whole.

C. Quantum Measurement and Bell Measurement

The projection postulate state that measurement irreversibly collapses the superposition state into one of its components [3]. Therefore, measurement on the qubit is a physical process that will disturb the original state. A projective measurement is

III. STANDARD QUANTUM TELEPORTATION PROTOCOL

IV. PHYSICAL MEANING AND EXPERIMENTAL RELEVANCE

As part of the activities in this course, you are asked to form groups of topical study, each of which consists of *at most five* people. Each group should pick one from the suggested topics provided in the “Notes 06 PHYS 7015 Syllabus 20260303.pdf” file. You are also welcome to propose other topics, and we will evaluate whether they are relevant to the course and at an appropriate level. Once the topic is selected/approved, you can proceed with the topical study with your teammates.

Your study can start from but *should not* be limited to the provided references. It is merely meant to be a starting point. Through literature search and study, we hope you will delve deeper into the topic and understand it better. Toward the end of the semester, each group is asked to make an oral presentation of about 40 minutes and answer questions from the audience, and to prepare a written report that should contain more details (derivations, discussions, figures, etc.) about your study.

The purpose of this activity is to train your ability in research, including literature survey, independent study, collaboration, professional writing, and oral presentation. You should digest the materials you have found, reorganize and present them in writing and in the oral report. You should play with practical numbers in the problem, generate a few plots, and so on, to help you gain more insight and understanding about the topic. It is good training about organizing your thoughts and presenting them in a logical and clear way so that not only can other readers easily understand and appreciate your report/presentation, but, more importantly, you can quickly remind yourself of what you have done/learned years later without difficulty.

The written report should be ready and shared with the class *before* your oral presentation.

V. ABOUT THE REPORT

A. Basic Structure of Report

Prepare a report *in* T_EX and *in* English that contains basically the following sections:

- introduction or motivations of the issue (e.g., what is the problem or question, why this topic is physically interesting, historical notes, etc.);

- model or framework for later discussions or analysis (e.g., defining notations, reviewing basic theory background, providing key equations, etc.);
- derivation in sufficient detail, leaving lengthy stuff to appendices, and discussions of physics in your results (e.g., deriving from the above-mentioned basics to the final results that will answer the initial question, providing more physical understanding about your results, discussing implications, comparing them with empirical data, etc.);
- summary and conclusions (e.g., reminding readers of the highlights in your study/report);
- using figures and tables whenever appropriate or necessary (in particular, you should make it clear whether the plots or numbers are computed by yourselves or copied from the references); and
- bibliography to cite essential and relevant references (learn from professional papers about how to do it properly).

B. About $\text{\LaTeX}$

C. Writing Style

You should write *in your own words* and provide *critical thoughts* on the topic.

Here are some additional tips about your writing:

- Make good use of modern technology to help you *check spelling and grammatical errors*. Pay attention to the proper usage of punctuation (including those in the equations).
- Be sure to clearly *define your symbols and notations*.
- Be *narrative* rather than simply showing a pile of equations. Provide a physical understanding of factors or terms in the derived results.
- Show sufficiently detailed steps in derivations, so that people at your level of understanding can follow without difficulty. In the case of tedious or trivial algebraic manipulations, you can schematically outline or describe how they are done.

- Role models would be most of the good textbooks and the research papers you have studied.

Appendix A: Useful Formulas

In the appendices, you can provide tedious derivations or detailed discussions that are not crucial for the discussions in the main text.

First, we have for $a > 0$

$$\begin{aligned}
G_0 &\equiv \int_{-\infty}^{+\infty} e^{-\frac{1}{2}ax^2} dx = \sqrt{\frac{2\pi}{a}} \\
G_2 &\equiv \int_{-\infty}^{+\infty} x^2 e^{-\frac{1}{2}ax^2} dx = \sqrt{\frac{2\pi}{a}} \frac{1}{a} \\
\text{or } G_{2n} &\equiv \int_{-\infty}^{+\infty} x^{2n} e^{-\frac{1}{2}ax^2} dx = \sqrt{\frac{2\pi}{a}} \frac{1}{a^n} (2n-1)(2n-3) \cdots 3 \cdot 1
\end{aligned} \tag{A1}$$

in general for $n \in \mathbb{N} \cup \{0\}$, where the factor of $1/2$ in the exponent is common in physics. The proof goes as follows:

$$\begin{aligned}
(G_0)^2 &= \left(\int_{-\infty}^{+\infty} e^{-\frac{1}{2}ax^2} dx \right) \cdot \left(\int_{-\infty}^{+\infty} e^{-\frac{1}{2}ay^2} dy \right) = \int_0^{2\pi} d\phi \int_0^\infty dr r e^{-\frac{1}{2}ar^2} \\
&= 2\pi \cdot \frac{1}{a} \int_0^\infty du e^{-u} = \frac{2\pi}{a} [-e^{-u}]_0^\infty = \frac{2\pi}{a} \\
\Rightarrow G_0 &= \sqrt{\frac{2\pi}{a}} \quad (\text{obviously not the other solution}) .
\end{aligned} \tag{A2}$$

The rest can be proved by induction. Suppose the last formula in Eq. (A1) is true for $n = k$, then for $n = k + 1$ we have

$$\begin{aligned}
\int_{-\infty}^{+\infty} x^{2(k+1)} e^{-\frac{1}{2}ax^2} dx &= \int_{-\infty}^{+\infty} x^{2k} x^2 e^{-\frac{1}{2}ax^2} dx = \int_{-\infty}^{+\infty} x^{2k} \left(-2 \frac{d}{da} \right) e^{-\frac{1}{2}ax^2} dx \\
&= \left(-2 \frac{d}{da} \right) G_{2k} = \sqrt{2\pi} (2k-1)(2k-3) \cdots 3 \cdot 1 \left(-2 \frac{d}{da} \right) a^{-(k+1/2)} \\
&= \sqrt{2\pi} (2k+1)(2k-1) \cdots 3 \cdot 1 a^{-[(k+1)+1/2]} .
\end{aligned} \tag{A3}$$

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- [1] M. A. Nielsen and I. L. Chuang, Quantum Computation and Quantum Information (Cambridge University Press, Cambridge, 2000), ISBN 0-521-63235-8.
- [2] C. H. Bennett, G. Brassard, C. Crépeau, R. Jozsa, A. Peres, and W. K. Wootters, Phys. Rev. Lett. **70**, 1895 (1993), URL <https://link.aps.org/doi/10.1103/PhysRevLett.70.1895>.
- [3] D. Bouwmeester, J.-W. Pan, K. Mattle, M. Eibl, H. Weinfurter, and A. Zeilinger, Nature **390**, 575 (1997), ISSN 1476-4687, URL <https://doi.org/10.1038/37539>.